

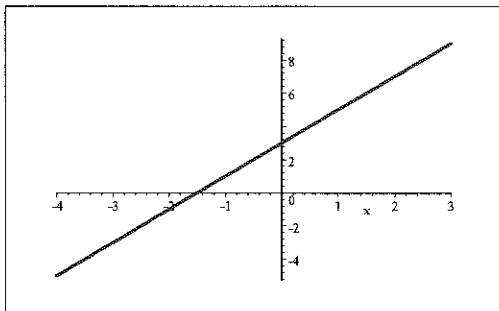
TECHNIQUE MASTERING EXERCISES

F - GRAPHS

SECTIONS C GRAPHS

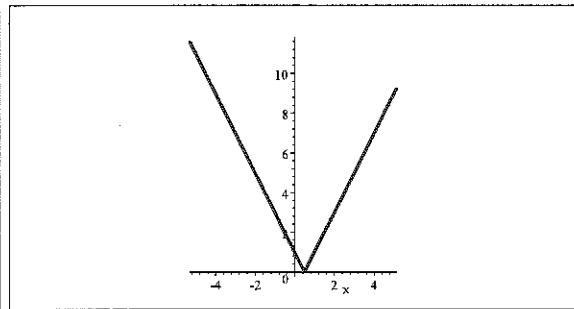
1.

a. $f(x) = 2x + 3$ on $[-4, 3]$



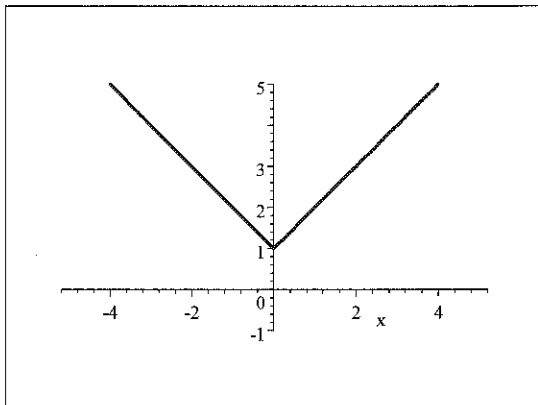
a.

b. $f(x) = |2x - 1| = \begin{cases} 2x - 1 & \text{if } x \geq \frac{1}{2} \\ -2x + 1 & \text{if } x < \frac{1}{2} \end{cases}$



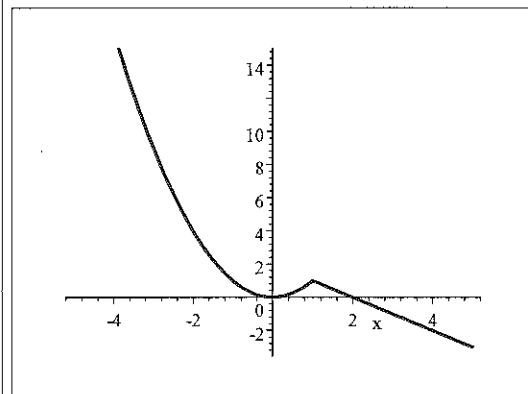
b.

c. $f(x) = |x| + 1$



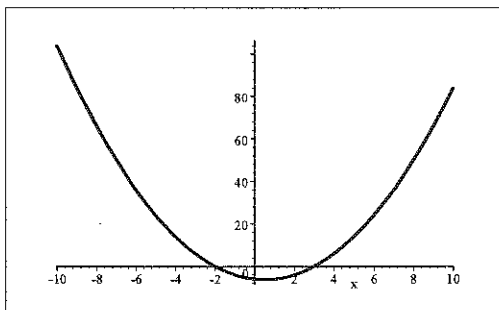
c.

d. $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ -x + 2 & \text{if } x > 1 \end{cases}$

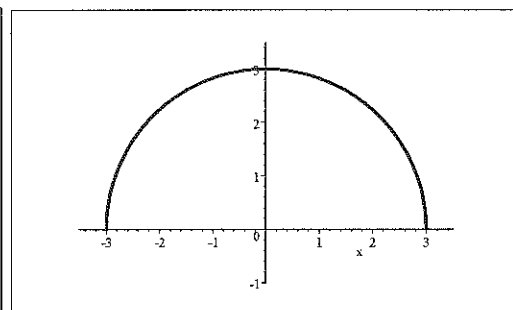


d.

e. $f(x) = x^2 - x - 6 = (x + 2)(x - 3)$ f. $f(x) = \sqrt{9 - x^2} \quad x \in [-3, 3]$

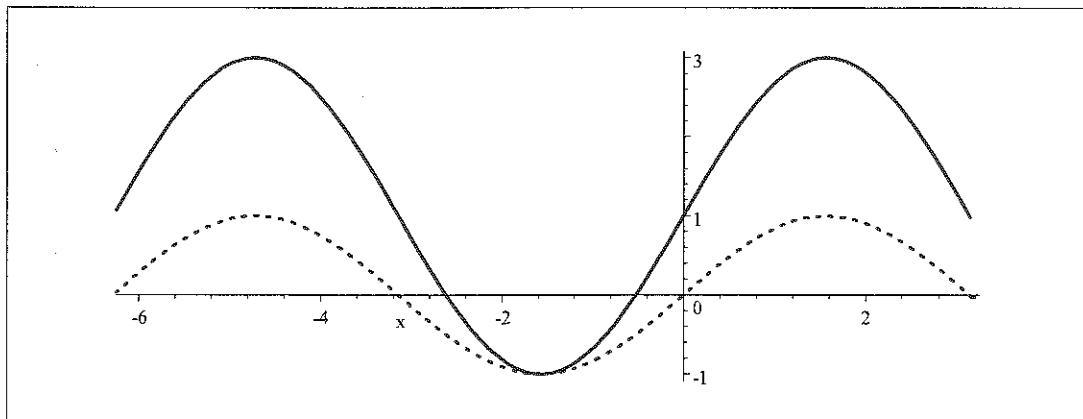


e.



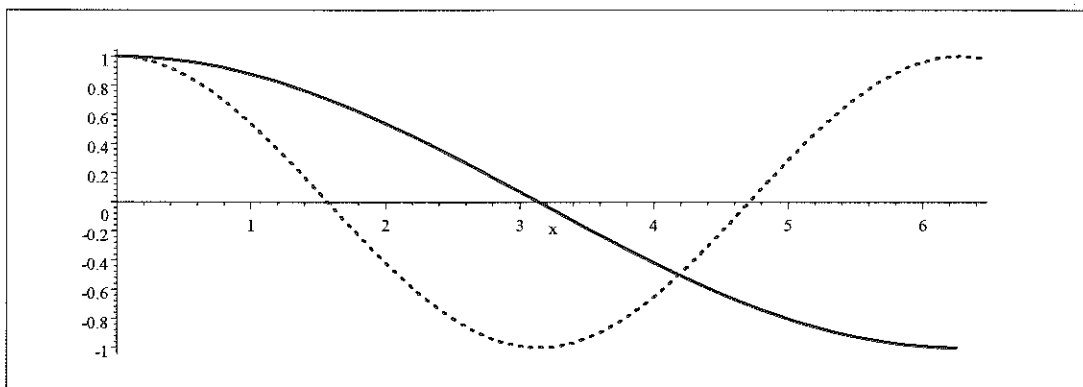
f.

g. $f(x) = 2 \sin x + 1$ on $[-2\pi, \pi]$,



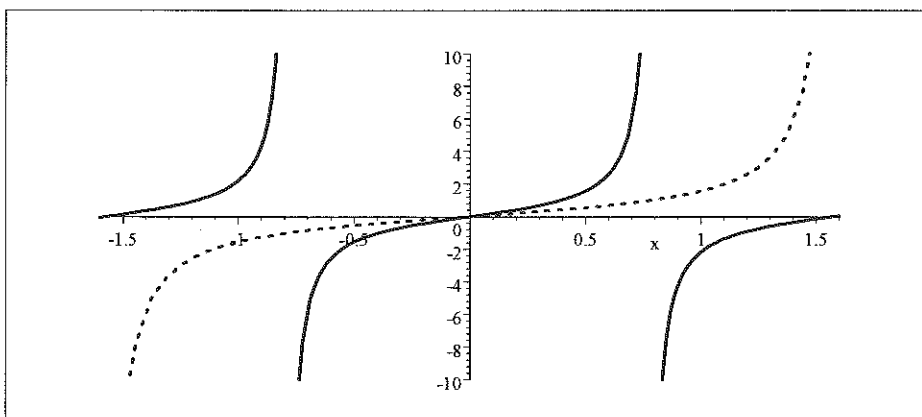
[f _____, $y = \sin x$...]

h. $f(x) = \cos \frac{x}{2}$ on $[0, 2\pi]$,



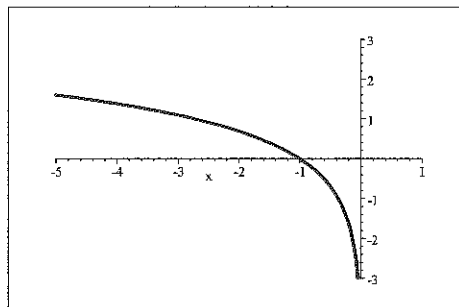
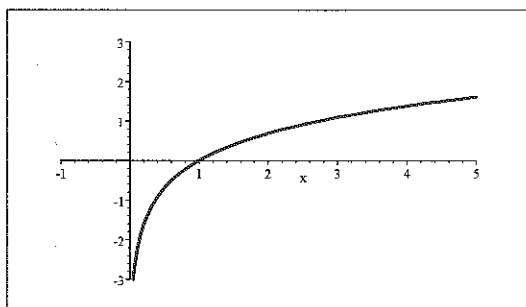
[f _____, $y = \cos x$...]

i. $f(x) = \tan 2x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$



[f _____, $y = \tan x$...]

2. a. $f(x) = \ln x, D_f = (0, \infty), W_f = \mathbb{R}$. b. $g(x) = \ln(-x), D_g = (-\infty, 0), W_g = \mathbb{R}$.

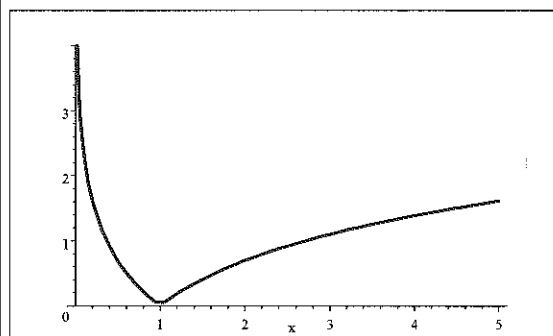
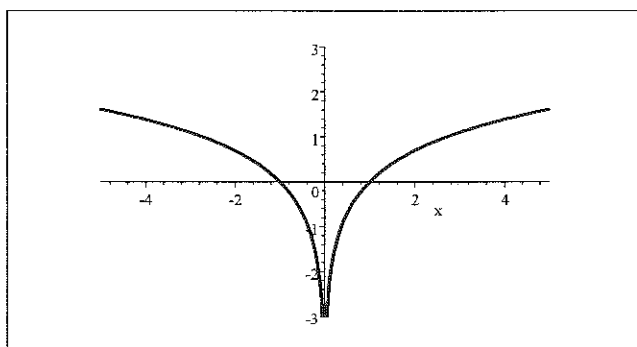


a.

b.

c. $h(x) = \ln|x| = \begin{cases} \ln x & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}, D_h = \mathbb{R} \setminus \{0\}, W_h = \mathbb{R}.$

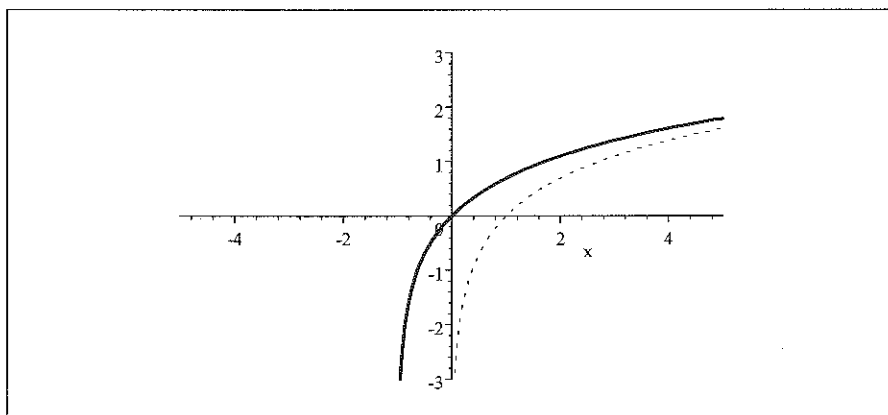
d. $l(x) = |\ln x| = \begin{cases} \ln x & \text{if } x \geq 1 \\ -\ln x & \text{if } 0 < x < 1 \end{cases}, D_l = (0, \infty), W_l = [0, \infty).$



c.

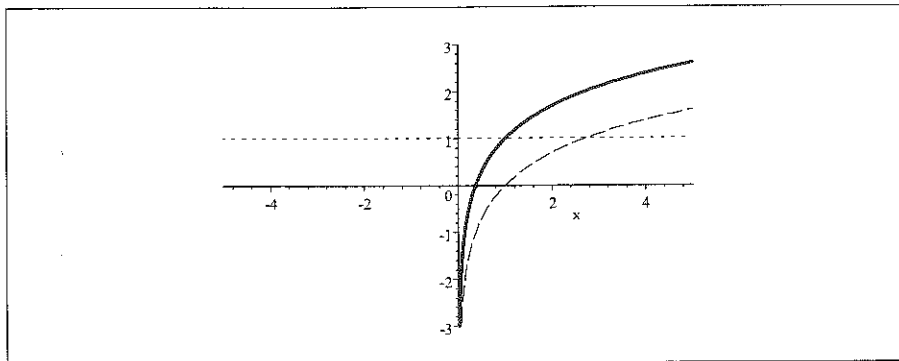
d.

e. $m(x) = \ln(1+x), D_m = (-1, \infty), W_m = \mathbb{R}.$



[m ____, y = ln x,]

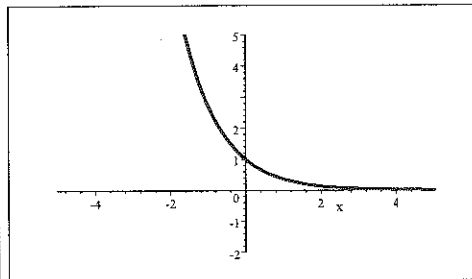
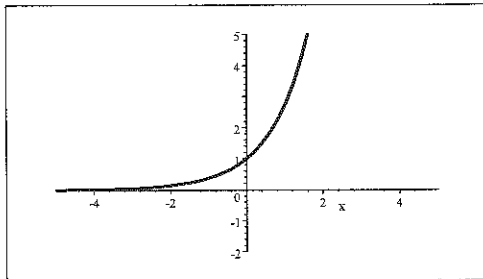
f. $n(x) = 1 + \ln x$ $D_n = (0, \infty)$, $W_n = \mathbb{R}$.



[m ____, $y = 1$..., $y = \ln x$ ----]

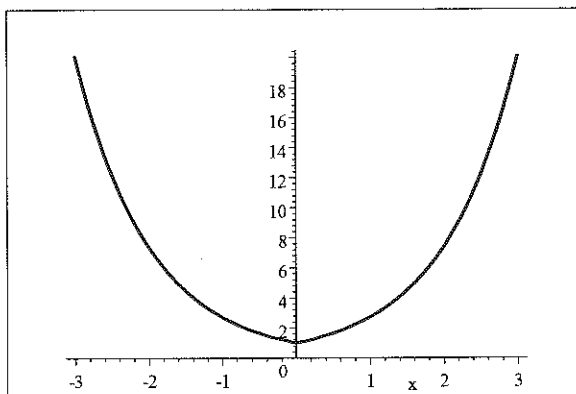
3.

a. $f(x) = e^x, D_f = \mathbb{R}, W_f = [0, \infty)$. b. $g(x) = e^{-x}, D_g = \mathbb{R}, W_g = [0, \infty)$.

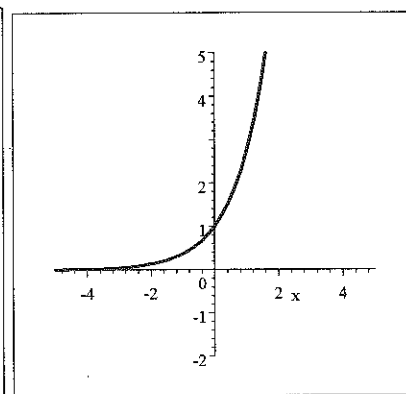


c. $h(x) = e^{|x|} = \begin{cases} e^x & \text{if } x \geq 0 \\ e^{-x} & \text{if } x < 0 \end{cases}, D_h = \mathbb{R}, W_h = [1, \infty)$.

d. $l(x) = |e^x| = e^x, D_l = \mathbb{R}, W_l = [0, \infty)$.

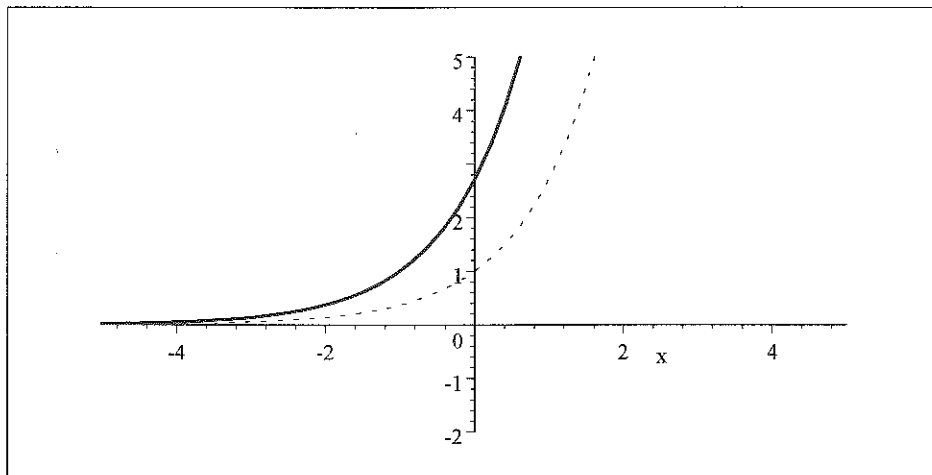


c.



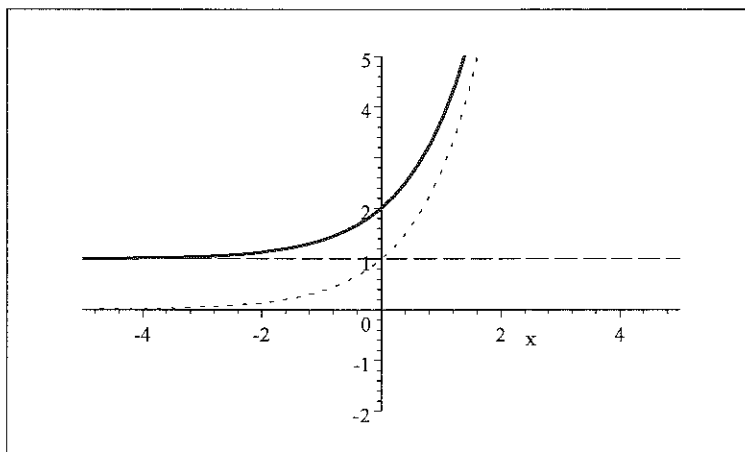
d.

e. $m(x) = e^{1+x}, D_m = \mathbb{R}, W_m = [0, \infty).$



[m _____, $y = e^x$ ]

f. $n(x) = 1 + e^x, D_n = \mathbb{R}, W_n = (1, \infty).$



[n _____, $y = 1$ ----, $y = e^x$ ]